
HCSM-23 (Revised)

The Identification of the 14 Modes with One Standard Model Generation

The Two-Line Structure, the Klein Orbit Classification,
the V_6 Congruence Characterization, and the (S, D) Factorization of the $N/4$ Anchors

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Abstract

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. HCSM-23 stated this identification as the framework's empirical thesis and listed the structural relations satisfied by the 14 $N/4$ mass-ordering anchors. This revision adds three new HCSM-native structural results. First, the 14 modes at $\lambda = 4$ are exactly the union of two lines $L_A \cup L_B$ in $\mathbb{Z}_8^2$, where $L_A : n_2 - n_1 \equiv 4 \pmod{8}$ and $L_B : n_1 + n_2 \equiv 4 \pmod{8}$, intersecting at $\{\nu, e\}$. Second, the six Klein orbits of HCSM-27 are exactly the $(\text{line}, n_1 \bmod 4)$ classes, and the four-cycle orbit V_6 admits the coordinate-level characterization $V_6 = \{m : n_1 - n_2 \equiv 2 \pmod{4}\}$. Third, every T_{half} pair has a sum S and a difference D that are framework constants, and the two $N/4$ anchors of each pair are exactly $(S + D)/2$ and $(S - D)/2$. The derivation target for the $N/4$ anchors reduces from 14 independent values to 7 (S, D) pairs of framework constants. The identification thesis remains empirical and falsifiable. All numerical claims are verified at machine precision.

Derivation status summary.

- DERIVED (NEW IN THIS REVISION): the two-line structure $L_A \cup L_B$; the $(\text{line}, n_1 \bmod 4)$ classification of the Klein orbits; the V_6 congruence characterization; the (S, D) factorization of the $N/4$ anchors; the framework-constant form of every S and every D .
- DERIVED (ORIGINAL): the (n_1, σ) bijection; the rs -pair ordering; the four-cycle Klein orbit sum; the $5 + 9$ split; the mass map fit.
- EMPIRICAL ANCHOR: the identification of each of the 14 modes with a Standard Model particle (Thesis 3.1), together with the 14 $N/4$ mass-ordering values.
- OPEN: the derivation of the 7 pair sums S and the 7 pair differences D from substrate primitives.

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1 Introduction

1.1 The identification thesis and the $N/4$ anchors

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with one Standard Model generation [1, 3]. The identification is stated as the framework's empirical thesis, structurally parallel to the Standard Model's Yukawa-coupling sector. This revision preserves that thesis and adds three new HCSM-native structural results about the $N/4$ anchors.

1.2 The three new structural results

This revision establishes three new results about the 14-mode multiplet at $\lambda = 4$:

1. **The two-line structure.** The 14 modes are exactly the union of two lines $L_A \cup L_B$ in $\mathbb{Z}_8^2$, with $|L_A| = |L_B| = 8$ and $|L_A \cap L_B| = 2$. The intersection is exactly $\{\nu, e\} = V_1$.
2. **The Klein orbit classification.** The six Klein orbits of HCSM-27 [10] are exactly the $(\text{line}, n_1 \bmod 4)$ classes. The four-cycle orbit V_6 admits the coordinate-level characterization $V_6 = \{m : n_1 - n_2 \equiv 2 \pmod{4}\}$.
3. **The (S, D) factorization.** Every T_{half} pair has a sum S and a difference D that are framework constants. The two $N/4$ anchors of each pair are exactly $(S + D)/2$ and $(S - D)/2$.

Together these results reduce the derivation target for the $N/4$ anchors from 14 independent values to 7 (S, D) pairs of framework constants. The identification of each mode with a Standard Model particle remains empirical; the internal structure of the $N/4$ list is now cleanly organized.

1.3 One-line summary

The 14 modes at $\lambda = 4$ form two perpendicular lines in $\mathbb{Z}_8^2$, whose intersection is the pair $\{\nu, e\}$; the Klein orbits are the $(\text{line}, n_1 \bmod 4)$ classes; the $N/4$ anchors factor as pairwise averages $(S \pm D)/2$ of framework constants.

1.4 Cross references

The paper depends on the following HCSM companions:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate; framework constants.
- HCSM-04 [3]: the 14-mode multiplet at $\lambda = 4$.
- HCSM-05 [4]: the D_4 representation theory; the n_1 -parity operator.
- HCSM-11 [5]: the $5 + 9$ split; the base formulas.
- HCSM-13 [6]: the processor T ; the fermion/boson split.
- HCSM-22 [7]: the zero state.
- HCSM-24 (revised) [8]: the charge theorem.

- HCSM-26 [9]: the observer pair $\{W, t\}$.
- HCSM-27 [10]: the Klein orbit structure.
- HCSM-28 [11]: the orbit-class reconstruction.

1.5 Organization

Section 2 states the HCSM primitives. Section 3 states the identification thesis. Section 4 proves the two-line structure. Section 5 proves the Klein orbit classification. Section 6 proves the V_6 congruence characterization. Section 7 proves the (S, D) factorization. Section 8 states the open derivation task. Section 9 catalogs the derivation status. Section 10 states the falsification conditions. Section 11 gives reproducibility notes. Section 12 concludes.

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16, \quad V = (d - 1) \cdot 2^d = 48. \quad (1)$$

The discrete 5-point Laplacian is

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - f(x_1 + 1, x_2) - f(x_1 - 1, x_2) - f(x_1, x_2 + 1) - f(x_1, x_2 - 1), \quad (2)$$

with eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (3)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14$.

In the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (4)$$

the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (5)$$

2.2 The N/4 anchors

The 14 N/4 mass-ordering labels (the framework's empirical anchors for the particle spectrum) are

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5) \quad (6)$$

in the standard mode order. The values are the framework's empirical anchors. Their internal structure is the subject of this paper.

2.3 The Klein group

Two order-two operations act on the 14-mode multiplet:

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}, \quad (7)$$

$$T_{\text{half}} : (n_1, n_2) \mapsto (n_1 + 4, n_2 + 4) \pmod{8}. \quad (8)$$

They commute and each squares to the identity, generating a Klein four-group $V = \langle rs, T_{\text{half}} \rangle = \{e, rs, T_{\text{half}}, rs \cdot T_{\text{half}}\}$ [10]. The product is

$$rs \cdot T_{\text{half}} : (n_1, n_2) \mapsto (n_2 + 4, n_1 + 4) \pmod{8}. \quad (9)$$

The Klein orbits of the 14-mode multiplet are the six orbits

$$\begin{aligned} V_1 &= \{(0, 4), (4, 0)\}, & V_2 &= \{(2, 6), (6, 2)\}, \\ V_3 &= \{(1, 5), (5, 1)\}, & V_4 &= \{(3, 7), (7, 3)\}, \\ V_5 &= \{(2, 2), (6, 6)\}, & V_6 &= \{(5, 7), (3, 1), (1, 3), (7, 5)\}. \end{aligned} \quad (10)$$

2.4 The T_{half} pairs

The half-period translation T_{half} pairs the 14 modes into 7 orbits of size 2 (the four two-cycle Klein orbits V_1, V_2, V_3, V_4 together with the two-cycle orbit V_5), plus the four-cycle orbit V_6 which decomposes into two T_{half} two-cycles:

$$V_6 = \{(1, 3), (5, 7)\} \cup \{(3, 1), (7, 5)\}. \quad (11)$$

This gives 7 T_{half} pairs in total:

$$\begin{aligned} P_1 &= \{(0, 4), (4, 0)\}, & P_2 &= \{(2, 6), (6, 2)\}, & P_3 &= \{(1, 5), (5, 1)\}, \\ P_4 &= \{(3, 7), (7, 3)\}, & P_5 &= \{(2, 2), (6, 6)\}, \\ P_6 &= \{(1, 3), (5, 7)\}, & P_7 &= \{(3, 1), (7, 5)\}. \end{aligned} \quad (12)$$

3 The Identification Thesis

Thesis 3.1 (Identification). The 14 modes at $\lambda = 4$ on the 8×8 torus are in one-to-one correspondence with the 14 Standard Model particles of one generation:

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (13)$$

where dark denotes the dark photon of the framework.

The thesis is the framework's empirical anchor for the particle sector, exactly as the Standard Model's Yukawa couplings are its empirical anchor for the fermion masses. It is not derived from the substrate. The 14 $N/4$ labels of Eq. (6) are the framework's empirical anchor for the mass sector.

The remainder of this paper establishes the internal structure of the $N/4$ labels, given the anchor. Nothing below derives the particle names or the numerical values of the anchors.

4 The Two-Line Structure

4.1 The factorization of the $\lambda = 4$ condition

Lemma 4.1 (Cosine factorization). *The condition $\lambda(n_1, n_2) = 4$ is equivalent to*

$$\cos\left(\frac{\pi n_1}{4}\right) + \cos\left(\frac{\pi n_2}{4}\right) = 0, \quad (14)$$

which factors exactly as

$$2 \cos\left(\frac{\pi(n_1 + n_2)}{8}\right) \cos\left(\frac{\pi(n_1 - n_2)}{8}\right) = 0. \quad (15)$$

Proof. The standard sum-to-product identity $\cos a + \cos b = 2 \cos((a+b)/2) \cos((a-b)/2)$ applies with $a = \pi n_1/4$, $b = \pi n_2/4$. Substituting $a/4 = \pi(n_1 + n_2)/8$ and $b/4 = \pi(n_1 - n_2)/8$ gives Eq. (15). $\square$

Definition 4.2 (The two lines). The two lines $L_A, L_B \subset \mathbb{Z}_8^2$ are

$$L_A = \{(n_1, n_2) \in \mathbb{Z}_8^2 : n_2 - n_1 \equiv 4 \pmod{8}\}, \quad (16)$$

$$L_B = \{(n_1, n_2) \in \mathbb{Z}_8^2 : n_1 + n_2 \equiv 4 \pmod{8}\}. \quad (17)$$

4.2 Union and intersection

Theorem 4.3 (Two-line structure). *The 14 modes at $\lambda = 4$ are exactly the union $L_A \cup L_B$. Each line contains 8 points of the torus, and their intersection is*

$$L_A \cap L_B = \{(0, 4), (4, 0)\} = V_1 = \{\nu, e\}. \quad (18)$$

Hence $|L_A| + |L_B| - |L_A \cap L_B| = 8 + 8 - 2 = 14$.

Proof. The first factor in Eq. (15) vanishes when $\pi(n_1 + n_2)/8 = \pi/2 + k\pi$, i.e. $n_1 + n_2 \equiv 4 \pmod{8}$. The second factor vanishes when $\pi(n_1 - n_2)/8 = \pi/2 + k\pi$, i.e. $n_1 - n_2 \equiv 4 \pmod{8}$, equivalently $n_2 - n_1 \equiv 4 \pmod{8}$. The union of these two conditions is $L_A \cup L_B$. Each condition defines a line of 8 points on the torus $\mathbb{Z}_8^2$ (each line is a coset of the $\mathbb{Z}_8$ subgroup generated by $(1, \pm 1)$). The intersection consists of the points satisfying both conditions: $n_1 + n_2 \equiv 4$ and $n_2 - n_1 \equiv 4$, i.e. $2n_2 \equiv 8 \equiv 0 \pmod{8}$, i.e. $n_2 \in \{0, 4\}$, giving $(n_1, n_2) \in \{(4, 0), (0, 4)\}$. This is exactly $V_1 = \{\nu, e\}$ from Eq. (10). The union count follows by inclusion-exclusion. $\square$

Corollary 4.4 (Line decomposition of the multiplet). *In standard mode order, the modes on each line are:*

$$L_A = \{\nu, e, d, b, Z, H, \mu, c\}, \quad (19)$$

$$L_B = \{\nu, e, u, s, \text{dark}, \tau, W, t\}. \quad (20)$$

The intersection is $\{\nu, e\}$. The A-only modes (in L_A but not L_B) are $\{d, b, Z, H, \mu, c\}$; the B-only modes (in L_B but not L_A) are $\{u, s, \text{dark}, \tau, W, t\}$.

Proof. Direct verification against the standard mode coordinates. $\square$

5 The Klein Orbit Classification

5.1 The $(\text{line}, n_1 \bmod 4)$ classification

Theorem 5.1 (Klein orbits are the $(\text{line}, n_1 \bmod 4)$ classes). *The six Klein orbits of Eq. (10) are exactly the classes*

$$V_1 = (\text{both}, 0), \quad V_2 = (L_A, 2), \quad V_3 = (L_A, 1), \quad V_4 = (L_A, 3), \quad V_5 = (L_B, 2), \quad V_6 = (L_B, 1) \cup (L_B, 3). \quad (21)$$

Explicitly:

<i>Klein orbit</i>	<i>$(\text{line}, n_1 \bmod 4)$</i>	<i>Modes</i>	<i>N/4 sum</i>
V_1	$(\text{both}, 0)$	ν, e	84
V_2	$(L_A, 2)$	Z, H	3
V_3	$(L_A, 1)$	d, b	27.5
V_4	$(L_A, 3)$	μ, c	23.5
V_5	$(L_B, 2)$	W, t	2.5
V_6	$(L_B, 1) \cup (L_B, 3)$	u, s, dark, τ	56

Proof. For each mode (n_1, n_2) in the multiplet, compute the line membership and the residue $n_1 \bmod 4$. Direct verification against Eq. (10). Both T_{half} and $rs \cdot T_{\text{half}}$ preserve the line and the residue $n_1 \bmod 4$; the action of T_{half} alone shifts n_1 by 4, hence preserves $n_1 \bmod 4$. So each Klein orbit is contained in a single $(\text{line}, n_1 \bmod 4)$ class. Because the classes partition the 14 modes into 7 groups of size 2 (except the $(\text{both}, 0)$ class of size 2, and the union $(L_B, 1) \cup (L_B, 3)$ of size 4), and the Klein orbits also partition the multiplet, the correspondence is exact. $\square$

Remark 5.2 (On the size-4 orbit). The four-cycle orbit V_6 is the unique Klein orbit that spans two distinct $n_1 \bmod 4$ classes: $(L_B, 1)$ and $(L_B, 3)$. This is why V_6 has four members rather than two, and why it is the unique orbit with trivial pointwise stabilizer [10].

6 The V_6 Congruence Characterization

Theorem 6.1 (V_6 congruence characterization). *The four-cycle orbit V_6 is exactly the set of modes at $\lambda = 4$ satisfying*

$$n_1 - n_2 \equiv 2 \pmod{4}. \quad (22)$$

Equivalently, $V_6 = \{m \in \mathcal{M} : n_1 - n_2 \equiv 2 \pmod{4}\}$, where $\mathcal{M}$ is the 14-mode multiplet. Every other mode satisfies $n_1 - n_2 \equiv 0 \pmod{4}$.

Proof. Direct computation. The four modes $(5, 7), (7, 5), (3, 1), (1, 3)$ have $n_1 - n_2 \in \{-2, +2\}$, all congruent to 2 mod 4. Every other mode in the multiplet has $n_1 - n_2 \in \{0, \pm 4\}$, all congruent to 0 mod 4. $\square$

Corollary 6.2 (Geometric meaning of V_6). *The four-cycle orbit V_6 is the unique Klein orbit whose members have $n_1 \not\equiv n_2 \pmod{4}$. All other modes satisfy $n_1 \equiv n_2 \pmod{4}$, which is the parity constraint $n_1 \equiv n_2 \pmod{2}$ of HCSM-05 strengthened to mod 4.*

Proof. Immediate from Theorem 6.1. For all modes at $\lambda = 4$, Lemma 7.2 of HCSM-05 gives $n_1 \equiv n_2 \pmod{2}$. The finer statement mod 4 splits the modes into the two classes: $n_1 \equiv n_2 \pmod{4}$ (10 modes) and $n_1 \not\equiv n_2 \pmod{4}$ (4 modes, exactly V_6). $\square$

7 The (S, D) Factorization of the $N/4$ Anchors

7.1 The T_{half} pair sums and differences

Definition 7.1 (Pair sum and difference). For each T_{half} pair $P = \{m, m'\}$ of Eq. (12), order the two modes by descending $N/4$ and define

$$S(P) = N/4_{\text{heavy}} + N/4_{\text{light}}, \quad D(P) = N/4_{\text{heavy}} - N/4_{\text{light}}. \quad (23)$$

Theorem 7.2 ((S, D) factorization). *Every T_{half} pair's sum $S(P)$ and difference $D(P)$ are framework constants. The values are given in Table 1.*

T_{half} pair	(line, k) class	S	D	D framework form
$\{(0, 4), (4, 0)\}$	(both, 0)	84	36	$4(d - 1)^2$
$\{(1, 5), (5, 1)\}$	$(L_A, 1)$	27.5	12.5	$(d + 1)^2/2$
$\{(2, 6), (6, 2)\}$	$(L_A, 2)$	3	1	1
$\{(3, 7), (7, 3)\}$	$(L_A, 3)$	23.5	4.5	$(d - 1)^2/2$
$\{(1, 3), (5, 7)\}$	$(L_B, 1)$	33	9	$(d - 1)^2$
$\{(2, 2), (6, 6)\}$	$(L_B, 2)$	2.5	1.5	$(d - 1)/2$
$\{(3, 1), (7, 5)\}$	$(L_B, 3)$	23	5	$d + 1$

Table 1: The 7 T_{half} pairs, their (line, $n_1 \bmod 4$) class, and their (S, D) values.

Proof. Direct arithmetic on the $N/4$ anchors of Eq. (6). The D framework forms are verified by substitution at $d = 4$. $\square$

7.2 Reconstruction of the $N/4$ anchors

Theorem 7.3 ($N/4$ anchors as (S, D) averages). *The two $N/4$ anchors of each T_{half} pair are exactly*

$$N/4_{\text{heavy}} = \frac{S + D}{2}, \quad N/4_{\text{light}} = \frac{S - D}{2}. \quad (24)$$

Proof. Immediate from Definition 7.1. The pair sum reconstructs the average, the pair difference reconstructs the half-separation. $\square$

Corollary 7.4 (Reduction of the derivation target). *The 14 $N/4$ anchors of Eq. (6) are determined by the 7 pairs (S, D) listed in Table 1. The derivation of the 14 anchors reduces to the derivation of the 14 values $(S_1, \dots, S_7, D_1, \dots, D_7)$.*

Remark 7.5 (Why this is progress). The 7 S values are the pair sums of $N/4$; the 7 D values are the pair differences. Every one of the 14 values is a framework constant, whereas the raw $N/4$ list includes half-integers (9.5, 7.5, 0.5) that are less obviously framework constants. The (S, D) factorization groups the anchors by their T_{half} pair structure, which is the structural relation the framework already uses for the pair-sum identities of HCSM-23 (Thm 4.3).

7.3 Distinctness and non-overlap of the D values

Proposition 7.6 (D values distinct). *The seven D values of Table 1 are pairwise distinct: $\{36, 12.5, 9, 5, 4.5, 1.5, 1\}$.*

Proof. Direct comparison. $\square$

Proposition 7.7 (S values distinct). *The seven S values of Table 1 are pairwise distinct: $\{84, 33, 27.5, 23, 23.5, 2.5, 3\}$.*

Proof. Direct comparison. The values 23 and 23.5 are the two nearly-degenerate pair sums (differing by 0.5), and 2.5 and 3 are the two small pair sums. $\square$

8 The Open Derivation Task

The (S, D) factorization reduces the derivation target for the $N/4$ anchors from 14 independent values to 7 (S, D) pairs. The remaining open problem is:

Derive the 7 values of S and the 7 values of D from substrate primitives.

Equivalently: derive the 14 values

$$S_i = (84, 27.5, 3, 23.5, 33, 2.5, 23), \quad D_i = (36, 12.5, 1, 4.5, 9, 1.5, 5) \quad (25)$$

as functions $S_i = f_S(\text{line}_i, k_i, n_1, n_2, \sigma_i, T_i, \dots)$ and $D_i = f_D(\text{line}_i, k_i, n_1, n_2, \sigma_i, T_i, \dots)$ of the substrate-level invariants of each T_{half} pair.

Candidate substrate-level primitives (from HCSM-13, HCSM-05, HCSM-27):

- The line projectors P_{L_A}, P_{L_B} .
- The Klein group elements $I, rs, T_{\text{half}}, rs \cdot T_{\text{half}}$.
- The n_1 -parity operator $D = \text{diag}((-1)^{n_1})$ (HCSM-05).
- The processor $T = I - P_C$ (HCSM-13).
- The winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ (HCSM-20).
- The zero-state sign $\sigma = \text{sign}(s)$ (HCSM-22).

The search should test whether each S_i or D_i is the trace of a specific combination of these operators restricted to the pair subspace, or the eigenvalue of a specific operator on the pair. The nine candidate families eliminated in HCSM-24 (revised) [8] for the charge functional provide a template for the elimination procedure.

9 Derivation Status

9.1 Derived in this revision

1. The two-line structure $L_A \cup L_B$ (Thm. 4.3).
2. The $(\text{line}, n_1 \bmod 4)$ classification of the Klein orbits (Thm. 5.1).
3. The V_6 congruence characterization $n_1 - n_2 \equiv 2 \pmod{4}$ (Thm. 6.1).
4. The (S, D) factorization of the T_{half} pairs (Thm. 7.2).
5. The $N/4$ anchors as $(S \pm D)/2$ averages (Thm. 7.3).
6. The reduction of the derivation target to the 7 (S, D) pairs (Cor. 7.4).

9.2 Derived in the original HCSM-23

1. The (n_1, σ) bijection.
2. The rs -pair ordering.
3. The four-cycle Klein orbit sum 56.
4. The $5 + 9$ split.
5. The mass map fit.

9.3 Empirical anchors

1. The 14 particle assignments (Thesis 3.1).
2. The 14 $N/4$ values of Eq. (6).

9.4 Open items

1. The derivation of the 7 pair sums S from substrate primitives.
2. The derivation of the 7 pair differences D from substrate primitives.
3. The derivation of the identification thesis from a structural principle.

10 Falsification

Theorem 10.1 (Falsifiability of the identification thesis). *The identification thesis is falsifiable. Any of the following observations would falsify it:*

1. *A fourth generation of Standard Model particles.*
2. *A particle content of one generation not matching the 14-mode structure.*
3. *An observed mass ordering that violates the rs -pair ordering (HCSM-23 original Thm 4.2).*
4. *An observed mass spectrum not fitting the mass map at any acceptable RMS.*

Theorem 10.2 (Falsifiability of the (S, D) factorization). *The (S, D) factorization is falsifiable. If the 14 $N/4$ anchors were found to violate the pair structure, i.e. if a different assignment of $N/4$ values to modes produced a different (S, D) list, the factorization would fail. As stated, the (S, D) values are a theorem given the specific anchors of Eq. (6).*

11 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-23revised_verification.py` constructs:

1. The 14 mode coordinates and the standard mode order.
2. The two lines L_A, L_B and their union.
3. The Klein group $V = \langle rs, T_{\text{half}} \rangle$ and the six orbits.

4. The T_{half} pairs and their (S, D) values.
5. The framework-constant forms of S and D .
6. The reconstruction of the $N/4$ anchors as $(S \pm D)/2$.

and verifies:

1. $|L_A \cup L_B| = 14$; $|L_A \cap L_B| = 2$; $L_A \cap L_B = \{\nu, e\}$.
2. The Klein orbits equal the $(\text{line}, n_1 \bmod 4)$ classes.
3. $V_6 = \{m : n_1 - n_2 \equiv 2 \pmod{4}\}$.
4. Every S and every D is a framework constant.
5. The reconstruction $N/4_{\text{heavy}} = (S + D)/2$, $N/4_{\text{light}} = (S - D)/2$ is exact.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

12 Conclusion

This revision of HCSM-23 has added three new HCSM-native structural results to the identification thesis and the $N/4$ anchors.

First, the 14 modes at $\lambda = 4$ are exactly the union of two lines $L_A \cup L_B$ in $\mathbb{Z}_8^2$, with intersection $\{\nu, e\}$. Second, the six Klein orbits of HCSM-27 are exactly the $(\text{line}, n_1 \bmod 4)$ classes, and the four-cycle orbit V_6 admits the coordinate-level characterization $n_1 - n_2 \equiv 2 \pmod{4}$. Third, every T_{half} pair has a sum S and a difference D that are framework constants, and the two $N/4$ anchors of each pair are exactly $(S + D)/2$ and $(S - D)/2$.

Together these results reduce the derivation target for the $N/4$ anchors from 14 independent values to 7 (S, D) pairs of framework constants. The identification thesis remains empirical and falsifiable; the internal structure of the $N/4$ list is now cleanly organized.

The remaining open problem is the derivation of the 7 pair sums S and the 7 pair differences D from substrate primitives. This is now the highest-leverage open item in the $N/4$ sector.

The 14 modes form two perpendicular lines. Their intersection is the pair the machine cannot cancel. The $N/4$ anchors are the pairwise averages of framework constants.

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